DiSC-AMC: Token- and Parameter-Efficient Discretized Statistics In-Context Automatic Modulation Classification

Mohammad Rostami, Atik Faysal, Reihaneh Gh. Roshan, Huaxia Wang, *Member, IEEE,* Nikhil Muralidhar, *Member, IEEE,* and Yu-Dong Yao, *Fellow, IEEE*

***Abstract*—The practical deployment of Large Language Models (LLMs) for in-the-loop Automatic Modulation Classification (AMC) is impeded by the bottlenecks of long prompt contexts and large model sizes. To address this challenge, we present Discretized Statistics in-Context Automatic Modulation Classification (DiSC-AMC), a token- and parameter-efficient variant that: (i) discretizes higher-order statistics and cumulants into compact symbolic tokens, (ii) prunes the exemplar list via a lightweight top-k neural prefilter and filters misleading/low-impact features using rationales extracted from prior LLM responses, and (iii) enforces label-only predictions through a calibrated prompt template. Together, these changes reduce both input/output tokens and the model parameter footprint by more than half while maintaining competitive accuracy. On synthetic AMC with ten modulation types under noise, a 7B *DeepSeek-R1-Distill-Qwen* baseline achieves 5.2% accuracy, whereas our system, using an approximately 5B-parameter *Gemini-2.5-Flash* model, attains 45.5% accuracy. These results demonstrate that careful discretization and context selection can cut inference cost by over 2× while preserving the advantages of prompt-based AMC and enabling practical in-the-loop use.**

***Index Terms*—Automatic modulation classification, large language models, prompt engineering, higher-order statistics.**

I. Introduction

AUTOMATIC Modulation Classification (AMC) is a pivotal technology in modern wireless communication systems, for applications like cognitive radio, spectrum sensing, and interference management. Accurate modulation identification is essential for efficient spectrum utilization and enhancing network adaptability and reliability. However, AMC remains challenging due to the effects of noise, interference, and channel impairments [1], [2].

Historically, AMC approaches evolved from traditional feature-based methods, relying on handcrafted signal features, to sophisticated deep learning models. Convolutional Neural Networks (*CNN*s) and, more recently, Transformer-based architectures, have demonstrated strong performance to achieve high classification accuracy, including in low Signal-to-Noise Ratio (SNR) environments [3]–[5]. Self-supervised denoising autoencoders further improve robustness and data efficiency under noise [6]–[8]. Despite these advancements, most deep learning solutions demand extensive labeled datasets and often require retraining or fine-tuning for new operating conditions, limiting their robustness and generalization.

Recent work advocates a Wireless Physical-layer Foundation Model (WPFM) for a general, adaptable backbone [2], [9]. This paradigm aligns with a broader trend of applying Large Language Models (LLMs) to structured, non-textual data, which often requires novel tokenization strategies to bridge the gap between continuous numerical data and the discrete nature of language models. For instance, LLM prompting expresses higher-order statistics as text to enable AMC via one-shot reasoning without LLM fine-tuning [10]. However, current LLM-based AMC is costly due to long numeric prompts and large models, limiting in-the-loop edge use.

We introduce Discretized Statistics in-Context Automatic Modulation Classification (DiSC-AMC), a framework for making LLM-based AMC both token- and parameter-efficient. Our core contribution is a redesigned prompt engineering pipeline that transforms continuous signal statistics into compact symbolic tokens, dynamically prunes in-context examples via a lightweight pre-filtering stage, and structures the prompt to enforce constrained, reliable decoding. This methodology cuts computational and token costs by more than half and enables smaller models to achieve performance competitive with larger ones, paving the way for practical real-world deployment.

II. Related Work

A. Automatic Modulation Classification

AMC literature has progressed from classical, feature-based classifiers [11] to supervised deep learning models that achieve state-of-the-art accuracy [3]–[6]. However, these deep learning solutions function as "closed-set" systems, requiring extensive training data and costly retraining to adapt to new modulation schemes or environments. Our work diverges from this paradigm by leveraging the In-Context Learning (ICL) ability of pre-trained LLMs, offering a path toward open-set generalization without the need for fine-tuning.

B. Tokenization for Scientific Data

A primary challenge in applying LLMs to scientific domains is tokenization, the conversion of continuous data into discrete symbols. A common baseline is to directly serialize floating-point values into text, as seen in the "plug-and-play" AMC approach [10]. Our work advances this by demonstrating that a carefully designed, coarse-grained symbolic discretization is not only more token-efficient but also improves performance by encouraging the LLM to engage in more abstract reasoning,

M. Rostami, A. Faysal, H. Wang are with the Department of Electrical and Computer Engineering, Rowan University, Glassboro, NJ, USA (e-mail: {rostami23, faysal24, wanghu}@rowan.edu).

R. Gh. Roshan and N. Muralidhar are with the Department of Computer Science (e-mail: {rghasemi, nmurali1}@stevens.edu), and Yu-Dong Yao is with the Department of Electrical and Computer Engineering (e-mail: yyao@stevens.edu). All authors are at the Stevens Institute of Technology, Hoboken, NJ, USA.

Code for this paper can be found at https://github.com/RU-SIT/DiSC-AMC

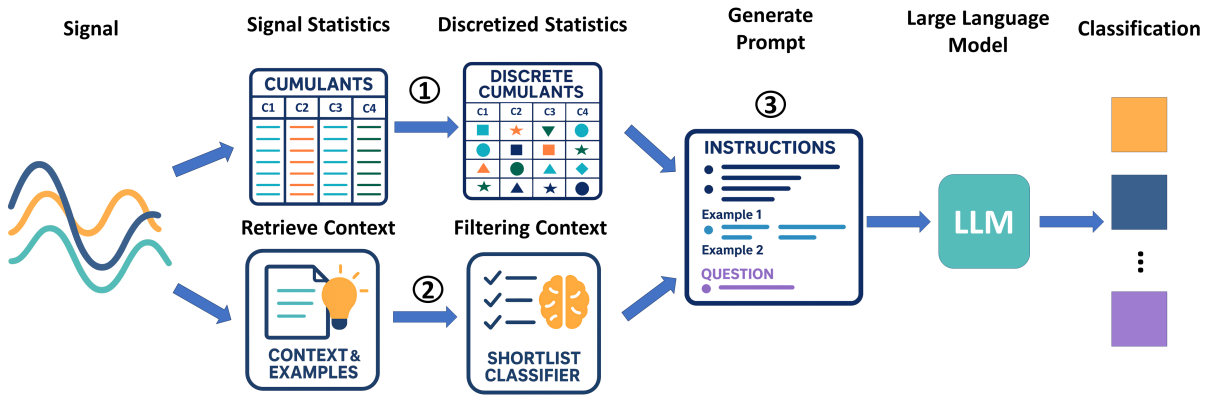


Fig. 1: Overview of the proposed DiSC-AMC framework. (1) Signal statistics such as cumulants (c1, c2, c3, ...) are computed and discretized into bins represented in the picture as different shapes. (2) Exemplars are pruned using a lightweight top-k shortlisting classifier. (3) The prompt is structured for efficient, constrained decoding by the LLM.

rather than getting lost in noisy, high-precision numerical details.

C. Efficient In-Context Learning

The performance of ICL is highly sensitive to the choice of exemplars. Much research has focused on developing better retrieval strategies, such as finding the most semantically similar examples to a query using *k-NN* [12], [13]. Our method introduces a fundamentally different approach. Instead of optimizing the selection of exemplars from a large pool, we use a swappable shortlisting module to radically prune the 'label space' itself. This creates a minimal, highly-targeted context that simplifies the reasoning task for the LLM. While we use a lightweight neural network, this module can be implemented with any efficient filter, such as deterministic rules or even another LLM. This strategy, combined with a constrained multiple-choice format, ensures the prompt is both efficient and reliable.

III. Method

We adopt a three-stage, plug-and-play pipeline adapted from a prior framework [10], redesigning each stage for improved efficiency. The pipeline involves: (1) discretizing I/Q signals into compact statistical tokens; (2) assembling a concise prompt using a pruned set of exemplars; and (3) reframing the query to enable constrained decoding. Fig. 1 illustrates the overall architecture of our proposed DiSC-AMC framework.

A. Stage 1: Discrete Statistical Tokens

We build upon the feature extraction technique of [10], computing descriptive statistics and cumulant-derived features from complex baseband segments. Our primary contribution is a novel discretization and tokenization scheme designed to optimize these features for LLMs. Whereas [10] serialized floating-point values directly, we map each scalar to a symbolic token corresponding to one of B discrete bins. This approach is crucial because it normalizes feature scales and compels the model to focus on qualitative patterns rather than irrelevant decimal details. Consequently, this symbolic representation significantly reduces the input token footprint, improves the robustness of in-context classification, and lowers overall inference costs.

B. Stage 2: Dynamic Prompt Pruning for Efficient Context

The effectiveness of ICL heavily depends on the quality and relevance of the provided examples. Using a large, fixed set of exemplars, as in prior work [10], is not only token-inefficient but can also introduce irrelevant information that degrades performance.

To address the token inefficiency and performance instability of ICL with large, static exemplar sets [10], we introduce a dynamic prompt pruning strategy. This method uses a lightweight visual classifier to create a compact, query-specific context. For each signal, the classifier analyzes its constellation diagram to identify the top-k most probable modulation classes [5]–[7]. The final prompt is then constructed using only the exemplars corresponding to this small, relevant subset, reframing the task as a constrained multiple-choice problem for the LLM.

The shortlisting classifier is built on a DINOv2 [14] Vision Transformer (*ViT-Base*) encoder and trained on a synthetic dataset of 10 modulation types across a -20 dB to +20 dB SNR range. As shown in Fig. 2, the classifier is highly effective; with $k = 5$, it achieves 99.83% accuracy, ensuring the correct class is almost always included in the candidate set provided to the LLM.

While our approach currently uses a pre-trained shortlisting classifier, the framework is readily adaptable to open-set and training-free scenarios. Open-set recognition can be enabled by simply incorporating an 'unknown' class into the prompt's options. Moreover, a fully training-free pipeline can be achieved by replacing the classifier with alternative shortlisting modules, such as deterministic functions or an agentic, LLM-based filter.

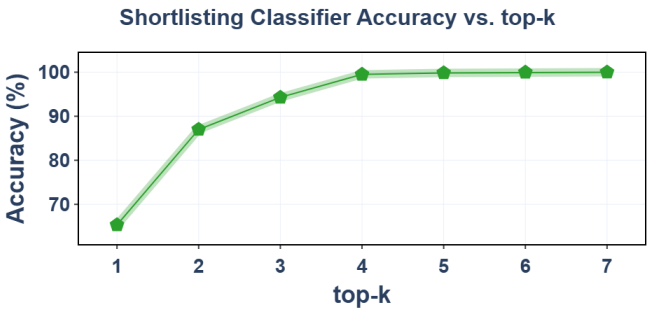


Fig. 2: Performance of the shortlisting classifier on top-k.

C. Stage 3: Improving Prompt Formulation with Constrained Decoding

Finally, we enhance reliability and enable constrained decoding through a structured prompt formulation. This process involves two key refinements. First, the query is reframed from an open-ended question into a multiple-choice format, providing the model with an enumerated list of valid class options, while the instruction block is improved with optimized templates. Second, the prompt's content is streamlined by removing low-impact statistical fields (e.g., *nobs/min/max/mean/variance*), which were identified using rationales extracted from prior LLM responses, and including the SNR, which helps replace long decimal strings with short, symbolic codes.

Although this detailed formulation slightly increases the raw prompt length, its structure facilitates more efficient inference. When combined with the token-saving measures from the preceding stages, this methodology yields a net efficiency gain of more than 2× in token and parameter usage while maintaining competitive accuracy. An example of this structured prompt is illustrated in Fig. 3.

```
(i) ROLE:
You are an expert AI signal classifier...

OBJECTIVE:
Your task is to classify the modulation scheme of
  a wireless signal based on...

CONTEXT:
The classification is based on the principle that
  moments and cumulants...
---
(ii) [EXEMPLARS]
Signal Statistics: snr: C, skewness: B,...
Classification Options: ['GMSK', ...]
Answer: OOK

... (top-$k$ pruned examples continue) ...
---
RESPONSE RULES:
1. MANDATORY: You MUST use `<think>` tags...
2. MANDATORY: After the closing `</think>`...
... (additional rules) ...
---
(iii) TASK EXECUTION:
Signal Statistics: snr: E, skewness: A, ...
Classification Options: ['DQPSK', ...]
Answer:
```

Fig. 3: A condensed example of the structured prompt used in DiSC-AMC. The prompt consists of (i) instructions defining the AI's role and task, (ii) a pruned set of in-context examples showing discretized signal statistics and their corresponding classifications, and (iii) a final query presented in a multiple-choice format to ensure constrained decoding.

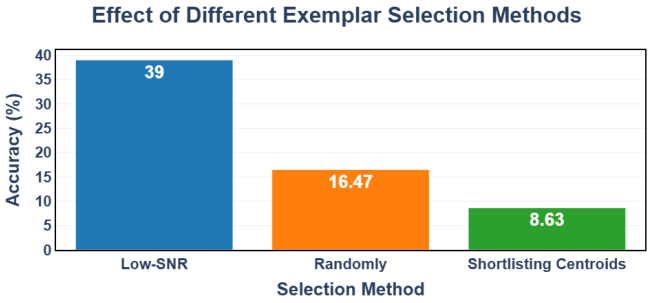


Fig. 4: Effects of different exemplar selection strategies on accuracy (*Gemini-2.5-Flash* [15], 10 bins, $k = 5$). The plot visualizes the instability of random selection versus the suboptimal performance of centroid-based selection.

IV. Experimental Setup

a) Dataset: We adopt the evaluation protocol from the plug-and-play [10] framework, following their methodology, and generate a new synthetic dataset. This dataset comprises I/Q signals representing 10 digital modulation types: 4ASK, 4PAM, 8ASK, 16PAM, CPFSK, DQPSK, GFSK, GMSK, OOK, and OQPSK. For each class, we generate 20 samples across an SNR range of -10 dB to +10 dB. All evaluations are performed in a one-shot, ICL setting where the model must classify a query signal given a single example of the selected class by the shortlisting classifier. Note that the dataset used in this work is newly generated and not identical to that of the original plug-and-play [10] paper; thus, results are not directly comparable across all methods in Table I.

b) Baselines: Our primary baseline is the plug-and-play framework [10], which prompts models with raw floating-point statistical features and a comprehensive, unpruned set of exemplars. To assess its performance, we apply this method to several open-weight models, including *DeepSeek-R1-Distill-Qwen-7B*, and *DeepSeek-R1-Distill-Qwen-32B*. Additionally, we report results from a larger, proprietary model (*o3-mini*) to establish a practical upper bound on performance. We also included results from other transformer-based models, including the Nmformer [5] and DenoMAE [6] for comparison.

c) Proposed Method and Models: We evaluate our three-stage pipeline, which integrates discretized statistical tokens, dynamic top-k exemplar pruning via a shortlisting classifier, and a structured multiple-choice prompt format. For our experiments, we use Google's Gemini models [15], accessed via their public API:

- *Gemini-2.5-Flash [15]:* A highly efficient model optimized for speed and low-cost inference.
- *Gemini-2.5-Pro [15]:* A state-of-the-art, high-performance model.

These models were selected not only for their diverse

positions on the performance-efficiency spectrum but also for their accessibility via a free public API, which facilitates reproducible research. Our primary metrics are classification accuracy across the SNR range and the final prompt length in tokens.

V. Experimental Results

Table I presents a comprehensive comparison of model performance and efficiency. It is important to note that the dataset used in this paper differs from that of the original plug-and-play [10] framework, so results are not directly comparable across all rows. The table clearly distinguishes between the baseline method, which uses unpruned, lengthy prompts (2.9K tokens), and our method, which uses compact, pruned prompts (e.g., 1.3K and 0.9K tokens), with the token count reflecting the number of selected exemplars (top-k) selected by the shortlisting classifier. The baseline method proves ineffective for smaller models; the 7B DeepSeek model achieves only 9% accuracy with the 2.9K token prompt. In stark contrast, our DiSC-AMC framework enables the even smaller 5B *Gemini-2.5-Flash* model to reach a competitive 45.50% accuracy using a prompt less than half the size (1.3K tokens). This result is particularly noteworthy, as it is comparable to the performance of the much larger 32B DeepSeek model using the baseline prompt (32.50%). Furthermore, when applying our method with the more powerful *Gemini-2.5-Pro* model, accuracy climbs to 51.00% with a highly efficient 0.9K token prompt, outperforming all other LLM-based configurations. While specialized supervised models like DenoMAE still hold an edge in absolute accuracy (81.30%), our approach offers the crucial advantages of being applicable to open-set and training-free scenarios without any LLM fine-tuning.

TABLE I: Accuracy and Efficiency Summary (Representative)

Model	Parameters	# Tokens	Accuracy (%)
Nmformer [5]	86M	-	71.60
DenoMAE [6]	86M	-	**81.30**
DenoMAE2.0 [7]	86M	-	82.40
DeepSeek-R1-Distill-Qwen [10]	7B	2.9K	05.20
DeepSeek-R1-Distill-Qwen [10]	32B	2.9K	47.80
o3-mini [10]	200B	2.9K	69.92
DeepSeek-R1-Distill-Qwen	7B	2.9K	09.00
DeepSeek-R1-Distill-Qwen	32B	2.9K	32.50
Gemini-2.5-Flash	5B	2.9K	29.50
Gemini-2.5-Pro	-	2.9K	42.50
DeepSeek-R1-Distill-Qwen (ours)	7B	1.3K	**33.50**
DeepSeek-R1-Distill-Qwen (ours)	32B	1.3K	**39.00**
Gemini-2.5-Flash (ours)	5B	1.3K	**45.50**
Gemini-2.5-Pro (ours)	-	0.9K	**51.00**

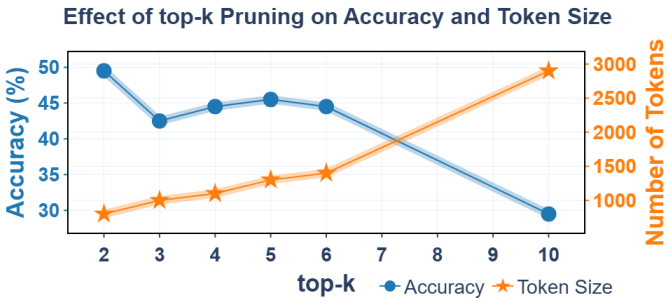


Fig. 5: Effect of prompt size (k) on accuracy and token count (*Gemini-2.5-Flash*, 5 bins). Increasing k shows diminishing returns, with a sharp performance drop for large contexts.

A. Ablation Studies

To better understand the contributions of individual components of our framework, we conduct a series of ablation studies.

1) Impact of Exemplar Selection: We investigated the critical impact of the exemplar selection strategy on model performance by comparing three distinct approaches, with results detailed in Fig. 4. Our analysis highlights the ineffectiveness of naive strategies. For instance, a deterministic method of selecting exemplars closest to class centroids proved suboptimal, yielding only 8.63% accuracy, likely because these samples lack the diversity needed for robust ICL. An alternative naive approach, random selection, also yields only 16.47%. This unreliability underscores the sensitivity of LLMs to the choice of in-context examples. In contrast to these methods, selecting exemplars with low SNR provides a more stable and effective solution. These findings collectively validate the need for a sophisticated pruning mechanism beyond simple heuristics to ensure reliable model performance. However, this is out of the scope of this study.

2) Effect of Prompt Size: Further ablations confirm the benefits of maintaining a compact prompt structure, as shown in Fig. 5. This experiment analyzes the effect of varying the number of exemplars (k) on accuracy and token count. We observe that increasing k from 4 to 5 provides only a marginal accuracy improvement (from 44.50% to 45.5%) while increasing the prompt length from 1.2K to 1.3K tokens. This indicates diminishing returns beyond a small number of carefully selected examples. More importantly, a significantly larger context, created by setting $k = 10$ and using 10 discretization bins, proves detrimental to performance. In this case, the prompt size balloons to 2.9K tokens, and the accuracy drops sharply to 29.50%. This result strongly supports our hypothesis that a concise, focused context is more effective than a large one that may contain distracting or irrelevant information.

3) Effect of Discretization Granularity: As shown in Fig. 6, the optimal discretization granularity is model-dependent. *Gemini-2.5-Flash* performance peaks at 45.5% with 5 bins and degrades monotonically with finer granularity. In contrast, *Gemini-2.5-Pro*'s performance is non-monotonic, peaking at 47.5% with 10 bins. The more capable Pro model appears to benefit from slightly more feature detail than the Flash model. Nevertheless, For both models excessively fine-grained features reduce accuracy, confirming that a tuned symbolic representation is superior to a high-precision one.

B. Complexity Analysis

1) Token Budget: Our framework achieves a substantial reduction in computational cost, primarily through a more efficient use of the token budget. As demonstrated in Table I, the baseline plug-and-play approach requires prompts exceeding 2,853 tokens. In contrast, our method, with its combination of

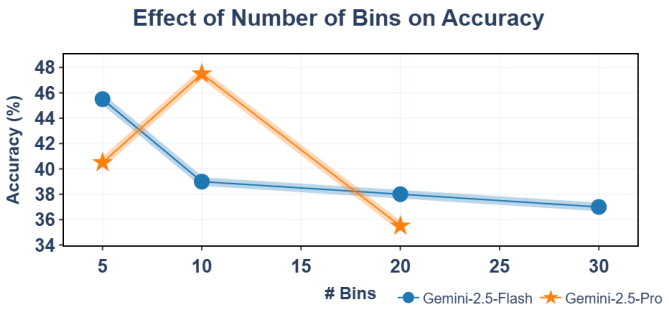


Fig. 6: Effect of discretization granularity on accuracy (k=5). Coarser quantization consistently leads to better performance in this noisy setting.

feature discretization and dynamic exemplar pruning, reduces this requirement to between 785 and 1315 tokens (Fig. 5), a decrease of over 50%. This efficiency stems from two key design choices: (i) reducing the number of statistical features from 21 floating-point values to 17 compact symbolic tokens, and (ii) pruning the number of in-context exemplars to a small, targeted set ($k \leq 5$), thereby minimizing redundant information and focusing the model's attention.

2) Parameter Budget: Beyond token efficiency, our approach enables the use of significantly smaller and more practical LLMs without a prohibitive loss in accuracy. Table I shows that our 5B *Gemini-2.5-Flash* [15] model achieves an accuracy of 45.5%, which is highly competitive with the 47.80% accuracy of the much larger 32B DeepSeek baseline. This represents an 84% reduction in model parameters, which translates directly to substantially lower Video Random Access Memory (VRAM) requirements and faster inference speeds. This dramatic reduction in the parameter budget makes in-context AMC feasible for deployment on resource-constrained hardware and edge devices, which is a primary goal of this work.

VI. Discussion

Our results show that compact LLMs are effective zero-shot AMC classifiers with careful prompt engineering. We find a 'less is more' principle applies: for noisy data, LLMs favor reasoning over abstract symbols rather than precise numerical inputs, as confirmed by the success of coarse discretization (Fig. 6) and compact contexts (Fig. 5).

The framework is adaptable: its shortlisting classifier is swappable (e.g., with training-free alternatives), and an 'unknown' prompt option enables open-set recognition. The paramount importance of prompt structure is highlighted by our 5B model's 45.5% accuracy, which far exceeds a 7B baseline's 9%. This results in an accuracy-efficiency trade-off practical for real-time applications. Future work will focus on adaptive feature selection and knowledge distillation to close the gap with specialized supervised models.

VII. Conclusion

We introduced DiSC-AMC, a token- and parameter-efficient framework that makes in-context AMC practical without LLM fine-tuning. By discretizing signal statistics into compact symbolic tokens and using a pruned, targeted prompt structure, our method cuts prompt length by over 50% and enables a 5B-parameter model to achieve accuracy competitive with a 32B-parameter baseline. These findings demonstrate a viable path toward deploying LLMs in resource-constrained wireless systems while preserving the benefits of open-set classification.

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